

Assignments

1. Modify the `adddigits` program so that it accepts a string which consists of both digits and non-digit characters. The program should however print the sum of only the digits, while ignoring the non-digit characters.

For example: if the string input is,
12ABC?3,
the sum is 6

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2. Write a SPIM program to encode the digits as shown below:

input digit: 0 1 2 3 4 5 6 7 8 9

output digit: 4 6 9 5 0 3 1 8 7 2

Your program should accept a string consisting of both digits and non-digits. The encoded string should be displayed in which only the digits are affected. Ask the user, whether he/she wants to terminate the program. If the response is “y” or “Y”, terminate the program; otherwise request for another input string.

Note that the above encoding has the property that if the encoding is applied twice, one gets the original string. Use this property to verify the program for ten arbitrary inputs.